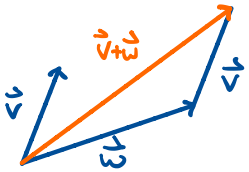
# 13.1 Displacement vectors

Vector $\vec{v}$ from P to Q.

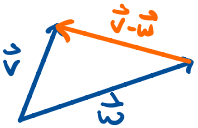
A **vector** from one point to another is an arrow with its tail at the first point and its head at the second. The **magnitude** of a vector is its length.

**Warning!** Vectors which have the same direction and magnitude are the same.

**Sum** $\vec{v}+\vec{w}$:

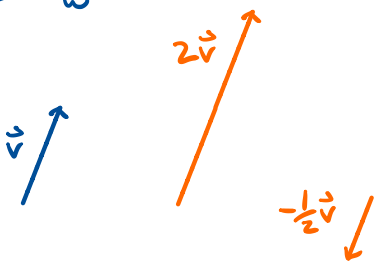



**Difference** $\vec{v}-\vec{w}$:

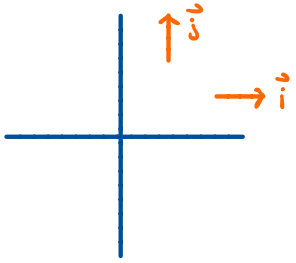



**Scalar multiple** $\lambda\vec{v}$:

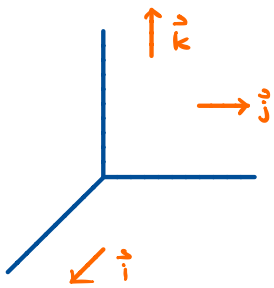
a number




## Basic vectors




Both $\vec{i}$ and $\vec{j}$ have length 1




All three of $\vec{i}, \vec{j}, \vec{k}$ have length 1

## Facts

- Every vector $\vec{v}$ can be resolved into the basis vectors as

$$\vec{v} = v_1\vec{i} + v_2\vec{j} \quad \text{in } \mathbb{R}^2$$

$$\vec{v} = v_1\vec{i} + \vec{v}_2\vec{j} + v_3\vec{k} \quad \text{in } \mathbb{R}^3$$

The $v_1, v_2, v_3$ are numbers.

→ Do problems 1 and 2.

- If $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$, the vector with tail at $P$ and head at $Q$ is

$$\overrightarrow{PQ} = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} + (z_2 - z_1)\vec{k}.$$

- If $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ and $\vec{w} = w_1\vec{i} + w_2\vec{j} + w_3\vec{k}$, then

$$\vec{v} \pm \vec{w} = (v_1 + w_1)\vec{i} + (v_2 + v_2)\vec{j} + (v_3 + w_3)\vec{k}$$

$$\lambda\vec{v} = \lambda v_1\vec{i} + \lambda v_2\vec{j} + \lambda v_3\vec{k}, \quad \lambda \text{ a number}$$

- The magnitude of $\vec{v}$ is

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

- A unit vector is a vector of length 1. A unit vector in the direction of $\vec{v}$ is $\vec{v}/\|\vec{v}\|$.

→ Do problems 3–5.